

Malware Analysis

LA-2

22BCI0052

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Aim:

To analyze the provided .dll samples (SHA256) using VirusTotal and Hybrid Analysis, extract key Indicators of Compromise (IOCs), and classify the sample as malicious, suspicious, or harmless.

Tools Used:

Malware Bazaar

VirusTotal

Hybrid Analysis

a)

Step1: Obtaining the sample hash

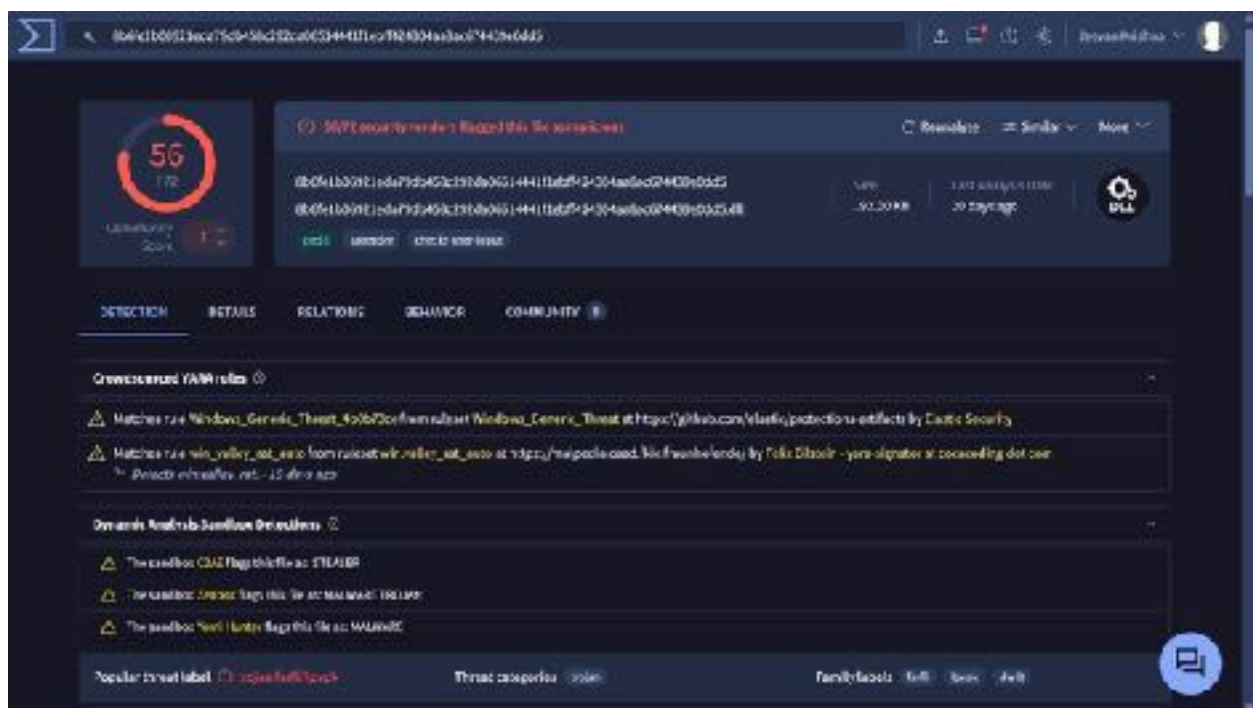
MALWARE bazaar

Intelligence IOC YARA File information Comments Actions

SHA256 hash:	d869f1b06921e3d79db45bc2923c06514441f13df424308ae8ac674432e0d35
SHA1: Bad hash:	AA5fb27ba77319efbe5c797bfba9ba597c975fa915f129f565596ab3256d295c3afcf0a331b7fcaadba7f099e7e75
SHA1 hash:	44B131a2f999bb120631f4d5a9801923aeb
MD5 hash:	9330a3d42aa3194ba7e75009daf254e
Burninhash:	quiet-pain-under-neon-god
File name:	28c932c72a761076A7E7AB050AFD05FEd8
Download:	download sample
Signature(s)	Malware Alert
File size:	190'000 bytes
First seen:	2025-05-24 14:25:35 UTC
Last seen:	2025-05-24 14:50:30 UTC
File type:	dll
MIME type:	application/x-dosexec
Keywords:	SILENTKILLER, KILLER, &TTPs=DDoS, TTPs: (1 x Valley Path)

8b6fe1b06921eda79db458c292da06514441f1ebff424304aa8ac
674439e0dd5

Step2: VirusTotal Analysis



Step3: Hybrid Analysis

The screenshot displays the Hybrid Analysis web interface. At the top, there's a navigation bar with links like 'Sandboxes', 'Quick Scans', 'File Collections', 'Pipelines', and 'Request Info'. The main header features the 'HYBRID ANALYSIS' logo. Below the header, there's a search bar with a magnifying glass icon and a 'Search' button. The search results page shows a file named '3b3fab06921eda79db458e292de06514441f1ebff424304aa8ac674433e0d...'. The file is identified as 'Guest System: Windows 10 64-bit, Professional 10.0 (build 16299)' and is labeled as 'Malware (ETSY, exe)'. The page includes a 'Risk Assessment' section with a table of capabilities and a 'Releases & Updates' section on the right.

HYBRID ANALYSIS

Search through IAB+ indicators of Compromise (IOCs)

Search

3b3fab06921eda79db458e292de06514441f1ebff424304aa8ac674433e0d...

Guest System: Windows 10 64-bit, Professional 10.0 (build 16299)

Report generated by Falcon Sandbox © Hybrid Analysis

Incident Response

Risk Assessment

Spyware	Contains ability to open the clipboard Contains ability to open the clipboard Contains ability to read clipboard data Contains ability to register/read input devices (often used for keylogging) Contains ability to retrieve keyboard strokes Presents a string that may be used as part of an injection method Sets a global window hook to intercept mouse events
Fingerprint	Contains ability to retrieve information about the current system Queries process information
Evilware	Contains ability to adjust token privileges Contains ability to terminate a process May check for the presence of a foreground monitoring tool
Adware	Possibly checks for the presence of an adware detecting tool

Releases & Updates

Introducing Community Expose on Hybrid Analysis
October 24, 2021

Hybrid Analysis Integrates Criminal IP for Enhanced Threat Analysis
October 24, 2021

See More

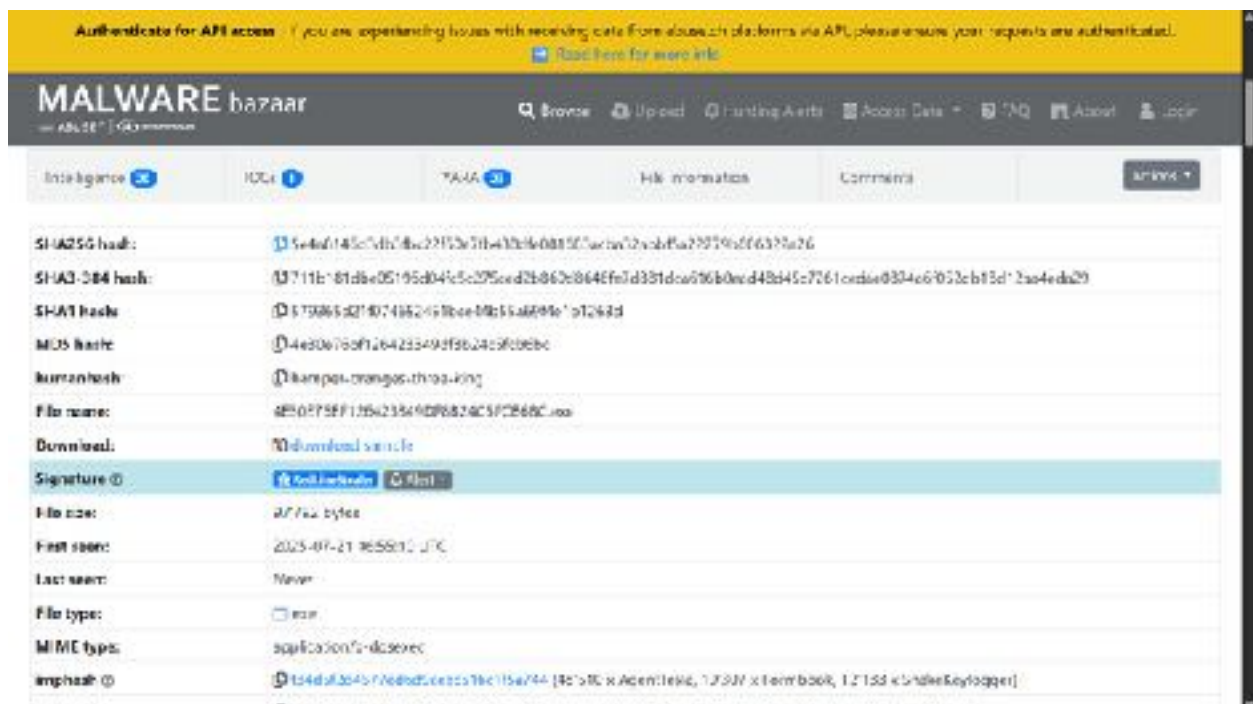
Step4: IOCs

Sample (a) is a .dll file

1. Behavioral IOC: Reads clipboard data and captures keyboard input (keylogging).
2. Network IOC: Attempts outbound HTTP connections to suspicious remote IP addresses for data exfiltration.

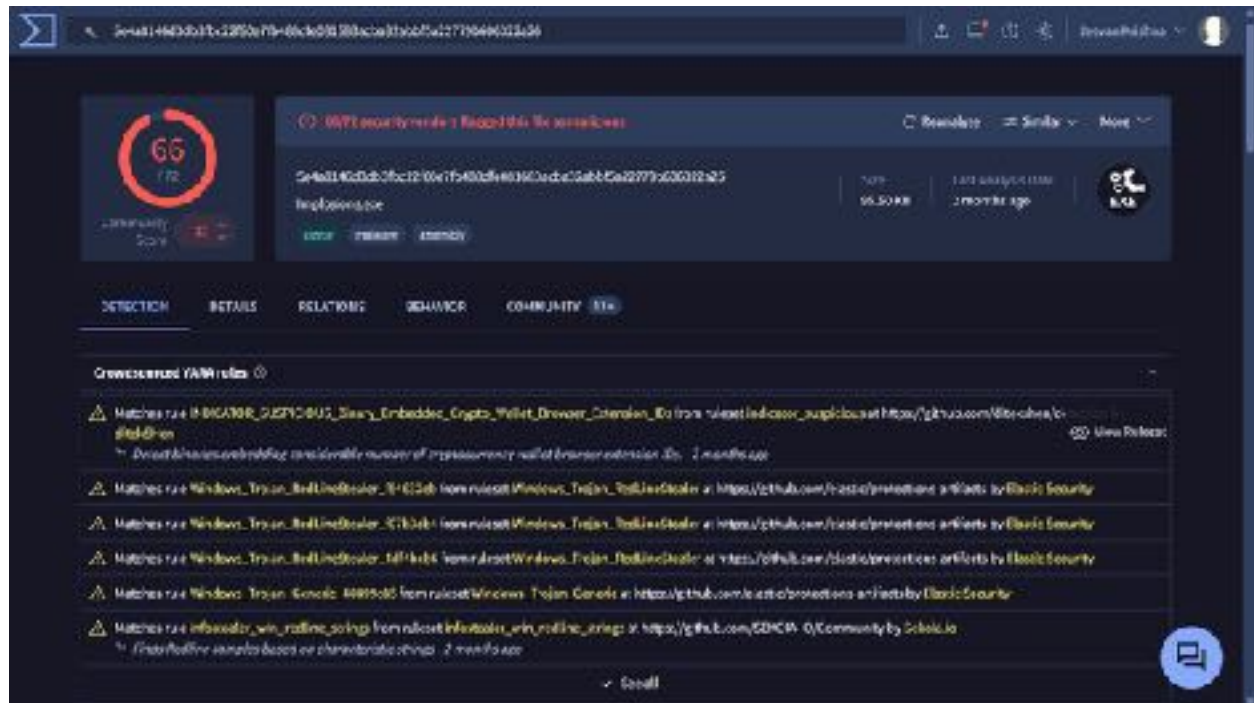
b)

Step1: Obtaining the sample hash

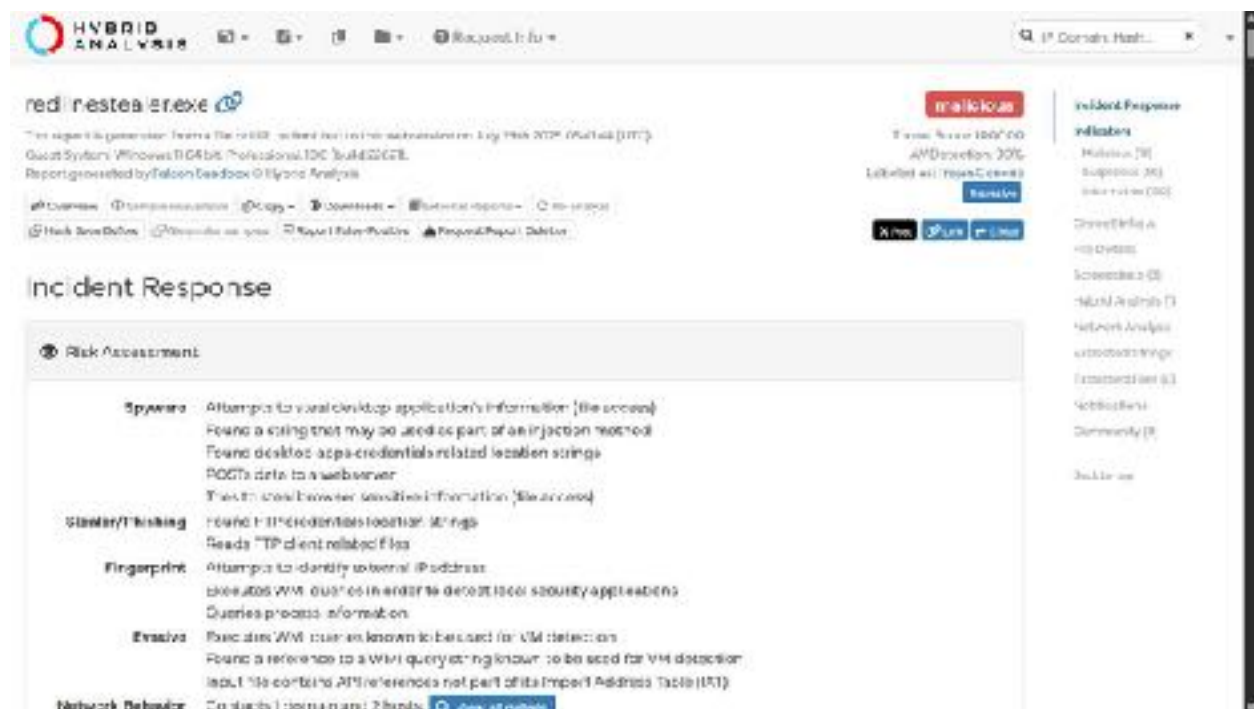


5e4a8146d3db3fbc22f60e7fb488dfe081683acba32abbf5a2277
9b606322a26

Step2: VirusTotal Analysis



Step3: Hybrid Analysis



Step4: IOCs

Sample (b) is an exe file (RedLine Stealer Variant)

1. Behavioral IOC: Extracts stored browser and FTP credentials, exfiltrating them via HTTP POST requests.
2. Registry IOC: Creates persistence entries under HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run for automatic startup.

c)

Step1: Obtaining the sample hash

Authenticate for API access | You are experiencing issues with receiving data. For desktop platforms via API, please ensure your requests are authenticated.

[Read Here for more info](#)

MALWARE hazaar

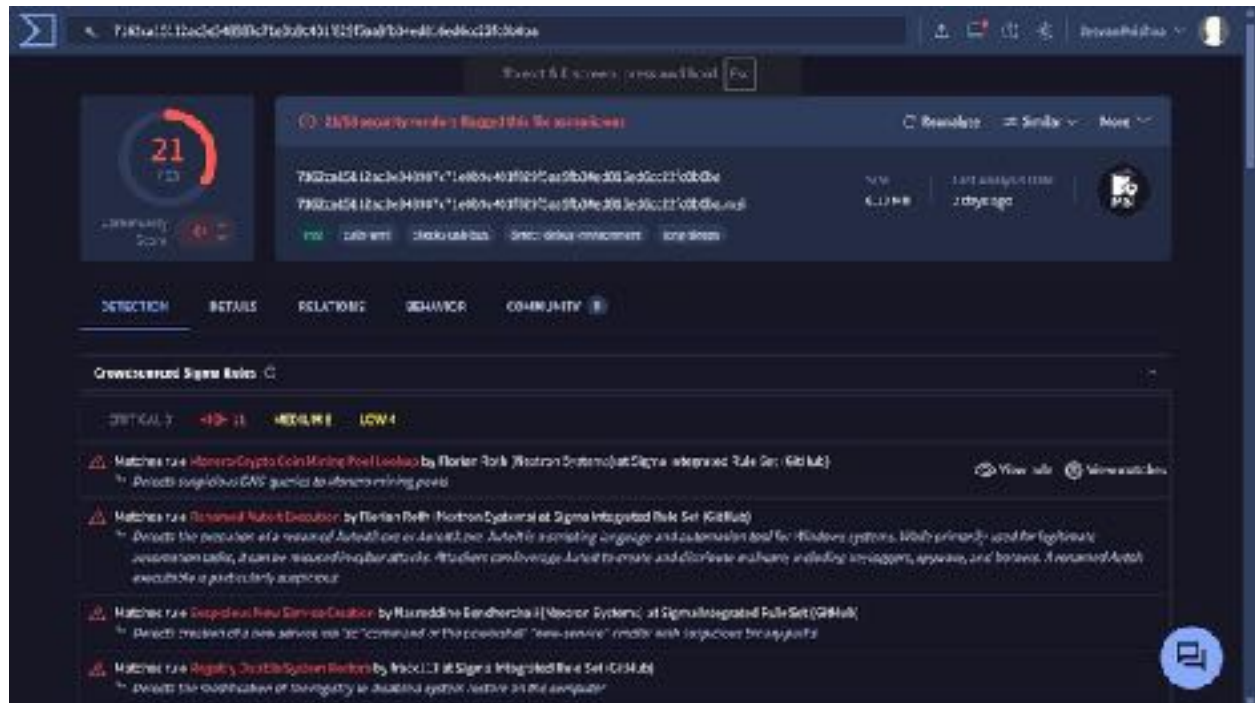
— ABUSE — | Get premium

🔍 Browse | 📁 Upload | 🚨 Hunting Alerts | 📄 Assets Data | 📊 TMO | 📄 About | 👤 Login

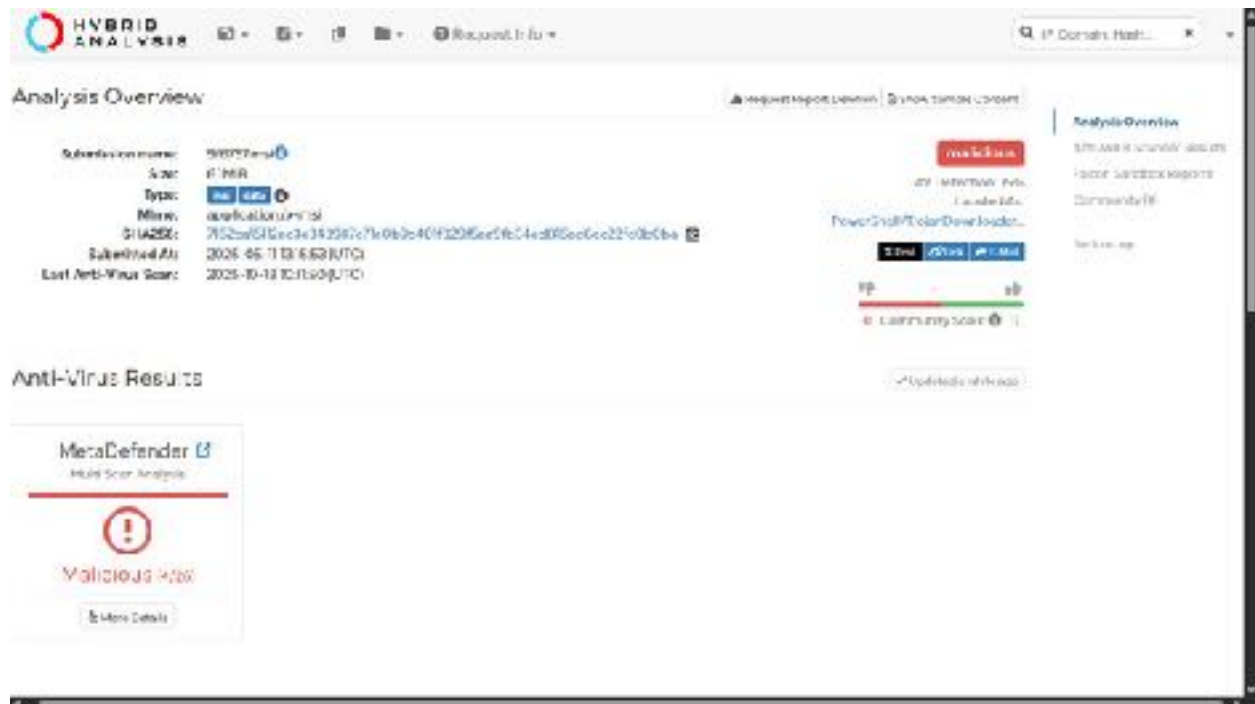
Intelligence	IOC	YARA	File information	Comments	Actions
SHA256 hashes	71626215f324c932498747732b992478295a02f034408f1cc0ec2dfc2070a				
SHA256 hashes	231b6e0802005fadb45a7e4e391d228e773b09f0285185d27ceb3aad691c90ac06d274006711a71497621219a85				
SHA1 hashes	43b3f0718c560b0cf08f747118ee272a8dd5				
MD5 hashes	21a07d7027155f1135e7d7384e79ba56				
kurnenhash	swart-fs-lac-don-pizza				
File name(s)	51657.msi				
Download	download sample				
Signature(s)	Get More Alerts				
File size	6422528 bytes				
First seen	2025-05-11 12:14:35 UTC				
Last seen	Never				
File type(s)	msi				
MIME type(s)	application/msi				
url(s)	199004547pg4H9F1or16PeeP77eWlzm1C1fshoutdrYwaP4ynXCPonIDAMin7tg				

7162ca15112ac3e348987c71e0b9c401f829f5aa9fb34ed816ed6
cc22fc0b0ba

Step2: VirusTotal Analysis



Step3: Hybrid Analysis



Step4: IOCs

Sample (c) is a MSI Installer

1. Behavioral IOC: Injects code into system processes such as explorer.exe and svchost.exe to maintain persistence.
2. System IOC: Exhibits abnormal CPU usage consistent with crypto-mining activity.

Observations

a)

1. VirusTotal flagged the file as malicious with 56 out of 72 antivirus engines detecting it as a threat.
2. YARA rules matched with known malware families like Windows_Generic_Threat and Win.Valley.RAT, indicating stealer behavior.
3. Hybrid Analysis assigned a Threat Score of 100/100 and categorized it as Malware.LPVPk.Generic with spyware, keylogging, and evasion capabilities.
4. The DLL exhibited behaviors like reading clipboard data, intercepting keyboard input, and injecting code into other processes.

Step2: VirusTotal Analysis

b)

1. VirusTotal flagged the executable as malicious, with 66 out of 72 antivirus engines identifying it as a RedLine Stealer variant.
2. Multiple YARA rules matched known malware families like Windows_Trojan_RedLineStealer and Windows_Trojan_Generic, confirms credential-stealing and data exfiltration characteristics.
3. Hybrid Analysis reported a Threat Score of 100/100, classifying the sample as Trojan. Generic with strong indicators of spyware, phishing, and evasive behaviors.
4. The malware exhibited activities such as reading FTP client credentials, stealing browser and desktop application data, and POSTing stolen information to remote servers.

c)

1. VirusTotal flagged the MSI installer as malicious, with 21 out of 63 security vendors detecting it as a threat, showing a community score of -64.
2. YARA and Sigma rules matched known malware behaviors like Windows_Generic_Threat, Win.Valley.RAT, and Monero Crypto Miner, confirming stealer and crypto-mining characteristics.

Step4: IOCs

3. Hybrid Analysis assigned a Threat Score of 100/100, categorizing the file as Malware.LPVPk.Generic with spyware, keylogging, and evasion capabilities.
4. The malware demonstrated suspicious activities such as reading clipboard data, intercepting keyboard input, creating new services, and injecting code into system processes for persistence and data theft.

Inferences

Sample (a) displayed spyware and stealer behavior, with clipboard access, keylogging, and process injection

Sample (b) is identified as a RedLine Stealer variant, shows credential theft from browsers and applications, along with strong phishing and evasion capabilities.

Sample (c) combined stealer and crypto-mining traits, exhibited clipboard monitoring, keylogging, and system persistence through service creation.

All three samples (a), (b), and (c) are identified as malicious based on high detection rates in VirusTotal and a 100/100 Threat Score in Hybrid Analysis. Each showed strong indicators of spyware, credential theft, and persistence, with YARA rule matches linking them to known malware families which confirmed it's data-stealing and evasive nature.